FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2022
FULL TEST – VIII
PAPER –2
TEST DATE: 14-08-2022

ANSWERS, HINTS & SOLUTIONS

Physics PART – I

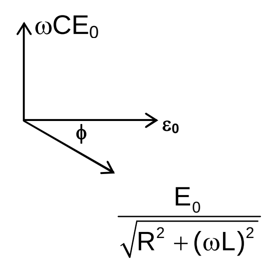
Section – A

1. B, C

Sol. $\frac{E_0 \sin\phi}{\sqrt{R^2 + (\omega L)^2}} = \omega C E_0$

Solving, we get

$I = \frac{\varepsilon_0 R}{R^2 + \omega^2 L^2} = \frac{C E_0 R}{L}$



2. A, C, D

Sol. $I = \frac{12}{16+8} = 0.5\ A$

$Q_{C_1} = 80\mu C$

$Q_{C_2} = 80\mu C$

No current is flowing through G is steady state.

3. A, D

Sol. Charge on capacitor is

$q = \frac{2CV}{3}\left(1 - e^{-\frac{3t}{2RC}}\right)$

$v_1 = \frac{2V}{3}\left(1 - e^{-\frac{3t}{2RC}}\right)$

$V_2 = V - \frac{2V}{3}\left(1 - e^{-\frac{3t}{2RC}}\right)$

When, $V_1 = V_2$, $t = \frac{4RC}{3}\ell n2$

4. A, C, D

Sol. $\frac{nh}{2\pi} = \frac{2h}{\pi} \Rightarrow n = 4$

$\frac{n^2}{z}a_0 = 4a_0 \Rightarrow z = 4$

$\frac{1}{\lambda_1} = Rz^2\left(1 - \frac{1}{4^2}\right) \Rightarrow \lambda_1 = \frac{1}{15R}$

$\frac{1}{\lambda_2} = Rz^2\left(\frac{1}{2^2} - \frac{1}{4^2}\right) \Rightarrow \lambda_2 = \frac{1}{3R}$

$\frac{1}{\lambda_3} = Rz^2\left(\frac{1}{3^2} - \frac{1}{4^2}\right) \Rightarrow \lambda_3 = \frac{9}{7R}$

5. A, C

Sol. $\vec{F}_b = -\rho_L V\vec{g}_{eff}$; $\vec{g}_{eff} = \vec{g} - \vec{a}$

$\vec{g}_{eff} = -g\hat{j} - a\hat{i}$

$\vec{F}_b = 4\rho V(a\hat{i} + g\hat{j})$

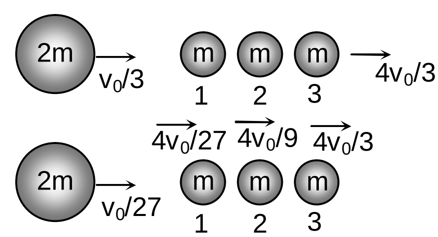
a = 2g

Also, $T = \sqrt{\frac{L}{3g}}$

6. A, C, D

Sol. When 2m collides with ball 1 which in turn with 2 and then with 3, then

And similarly in final state



7. A

8. B

Sol. **(for Q.7-8)**

$\lambda_m T = b \Rightarrow \ell n\lambda_m = \ell nb - \ell nT$

Here $\ell nb = A \Rightarrow b = e^A$

9. D

10. C

Sol. **(for Q. 9-10):**

For mass

(M + m)a = (M + m)g – T …(i)

For mass

T – Mg = Ma …(ii)

From (i) and (ii)

$a = \frac{mg}{2M + m}$

For cork with respect to bucket

T′ + mg = ma + B

$$T' = \frac{2mMg}{(m+2M)}\left(\frac{1}{\sigma}-1\right)$$

Section – B

11. 9

Sol. $\tan\theta = \frac{a_0}{a_t} = \frac{\omega^2 R}{\beta R} = \frac{\omega^2}{\beta}$

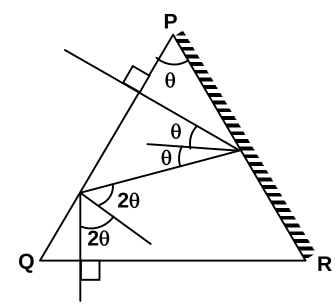
$$\tan\theta = \frac{\left(at^3/3\right)^2}{at^2} = \frac{at^4}{9}$$

$$\Rightarrow t = \left(\frac{9\tan\theta}{a}\right)^{1/4}$$

$\therefore \alpha = 9$

12. 12

Sol. $\left(\frac{\pi}{2}-\frac{\theta}{2}\right)+\left(\frac{\pi}{2}-2\theta\right) = \frac{\pi}{2}$



$$\Rightarrow \frac{5\theta}{2} = \frac{\pi}{2}$$

$$\theta = \frac{\pi}{5} = 36^\circ$$

$$\therefore \frac{\theta}{3} = 12^\circ$$

13. 60

Sol. For mirror formula

$$\frac{1}{v}+\frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{20/3}+\frac{1}{-20} = \frac{1}{f} \Rightarrow f = 10 \text{ cm}$$

Also, $-\frac{1}{f} = \frac{2}{f_e} - \frac{1}{f_m}$

$$\Rightarrow -\frac{1}{10} = \frac{2}{30} - \frac{1}{f_m}$$

$\Rightarrow f_m = 30$ cm

$\therefore$ R= 60 cm

Section – C

14. 7.00

Sol. Let $V_1 = 2V_0$ and $V_2 = 3V_0$

$C_1 = C$ and $C_2 = 3C$

When current is maximum then at that instant potential difference across L is zero. Hence potential difference across C_1 and C_2 is same.

$CV + 3CV = 9CV_0 - 2CV_0$

$$V = \frac{7V_0}{4} = \frac{7}{4}(4) = 7 \text{ volt.}$$

15. 1000.00

Sol. From conservation of energy,

$$\frac{1}{2}LI_{max}^2 = \left[\frac{1}{2}(2v_0)^2 + \frac{1}{2}3C(3v_0)^2\right] - \frac{1}{2}(C+3C)V^2$$

Here V = $\frac{7V_0}{4}$

Solving we get,

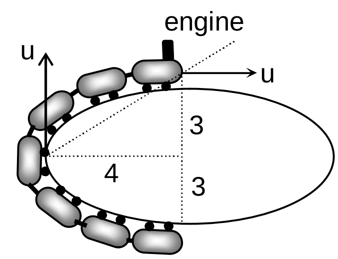
$$I_{max} = \frac{5V_0}{2}\sqrt{\frac{3C}{L}} = \frac{5}{2}(4)\sqrt{\frac{3\times10\times10^{-6}}{3\times10^{-3}}} = 1 \text{ A} = 1000 \text{ mA}$$

16. 1730.00

17. 1710.00

Sol. **(for Q. 16-17):**

$$f = 1730\left[\frac{330+20\times\frac{3}{5}}{330+20\times\frac{4}{5}}\right] = 1710$$



18. 1.50

Sol. $A = \int_{\pi/3}^{\pi/2} 2\pi(R\cos\theta)(Rd\theta) = \left(1-\frac{\sqrt{3}}{2}\right)2\pi R^2$

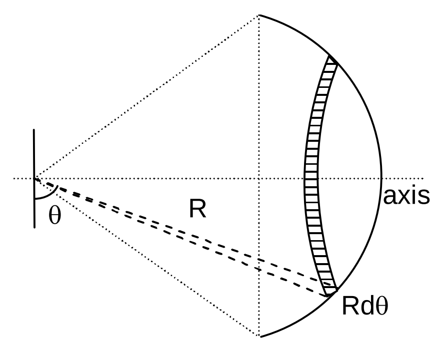
Area of both caps

$2A = (2-\sqrt{3})2\pi R^2$

$\therefore$ mass of leftover part is

$$m' = \frac{M - m(2-\sqrt{3})2\pi R^2}{4\pi R^2}$$

$$= \frac{\sqrt{3}}{2}M = 1.5 \text{ kg}$$



19. 18.00

Sol. $I = \int_0^{\pi/3} 2\cdot\frac{M}{4\pi R^2}\cdot 2\pi R^2\cos\theta(R\cos)^2 d\theta$

$$= MR^2\int_0^{\pi/3}\cos^3\theta = \frac{3\sqrt{3}}{8}MR^2$$

$$I = \frac{3\sqrt{3}}{8}(\sqrt{3})(4)^2 = 18 \text{ kg-m}^2$$

Chemistry

PART – II

Section – A

20. A, B

Sol.
$$HNO_2 + NH_2 - CO - NH_2 \longrightarrow \underset{(X)}{N_2} + \underset{(Y)}{CO_2} + \underset{(Z)}{H_2O}$$

$$\underset{(X)}{Mg + N_2} \longrightarrow \underset{(W)}{Mg_3N_2} \xrightarrow{H_2O} \underset{(P)}{NH_3} \xrightarrow{CuO} \underset{(X)}{N_2}$$

21. A, B, C, D

Sol. Ti^{2+}, V^{2+}, Cr^{2+} acts as reducing agent

$Mn^{3+}, Fe^{3+}, Co^{3+}$ acts as oxidizing agent.

22. A, B, D

Sol. $CH_3 - CH_2 - CN(P)$

$CH_3 - CH_2 - CH_2 - NH_2\ (Q)$

23. A, C

Sol. Hoffman elimination.

24. A, B, D

Sol.
$$Ag + NaCN + O_2 \longrightarrow Na\left[Ag(CN)_2\right] + Na_2SO_4$$

$$Zn + 2\left[Ag(CN)_2\right]^- \longrightarrow 2Ag\downarrow + \left[Zn(CN)_4\right]^{2-}$$

25. B, C

Sol. $n_{O_2} = \frac{1}{2}$ $n_{O_3} = \frac{1}{3}$

Molecules of $O_2 = \frac{1}{2}N_A$, Molecules of $O_3 = \frac{1}{3}N_A$

Atoms in bulb A = N_A, Atoms in bulb B = N_A

26. B

Sol.
$$Y_A - X_A = \frac{X_A.P_A^o}{P_B^o + X_A.\left(P_A^o - P_B^o\right)} - X_A$$

$$= \frac{X_A\left(P_A^o - P_B^o\right) - X_A^2\left(P_B^o - P_A^o\right)}{P_B^o + X_A.\left(P_A^o - P_B^o\right)}$$

$$\frac{d\left(Y_A - X_A\right)}{dX_A} = 0$$

$$\therefore X_A = \frac{\sqrt{P_A^o.P_B^o} - P_B^o}{P_A^o - P_B^o}$$

27. A

Sol. $P_{Total} = P_B^o + X_A.\left(P_A^o - P_B^o\right) = \sqrt{P_A^o.P_B^o}$

28. C

Sol. Net cell reaction is independent from $OH^-(aq)$.

29. B

Sol. $-\Delta G^o = nFE_{cell}^o = 2\times 96500\times 1.27 = 245110$ J

Section – B

30. 72

Sol. $P_4 + 20HNO_3 \longrightarrow 4H_3PO_4 + 20NO_2 + 4H_2O$

31. 6

Sol. i = 1.25

original mole fraction = $\frac{1}{n} = \frac{1}{1+(n-1)}$

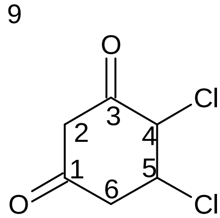
Now $\frac{1.25}{1.25+(n-1)} = \frac{1}{5}$

6.25 = 1.25 + (n – 1)

n = 6

32. 9

Sol.



Section – C

33. 2.00

Sol. $n_{eq}(KMnO_4) = n_{eq}(X^{n+})$

$1.5\times 10^{-3}\times 5 = 2.5\times 10^{-3}\times (5-n)$

$n = 2$

34. 3.00

Sol. $n_{eq}(KMnO_4) = n_{eq}(X^{n+})$

$n\times 1 = 1\times 3$

$n = 3$

35. 25.00

36. 37.50

Sol. **(for Q. 35 to 36):**

For zero order kinetics

$x = kt$

$t_{100\%} = \frac{a}{k} = 50 \text{ sec}$

$t_{50\%} = \frac{a}{2k} = 25 \text{ sec}$

$t_{75\%} = \frac{3a}{4k} = 37.5 \text{ sec}$

37. 5.00

38. 2.00

Sol. **(for Q. 37 to 38):**

Total stereoisomers $= 2^{n-1}$

$= 2^{3-1} (n = 3)$

$= 4$

Optically inactive $= 2^{\frac{n-1}{2}}$

$= 2$

Optical active = 4 – 2 = 2

Number of diastereomeric pair = 5

Mathematics PART – III

Section – A

39. A, C, D

Sol. $(1 + x + x^2)^n = a_0 + a_1x + a_2x^2 + ...$ ..... (1)

$(1 - x)^n = {}^nC_0 - {}^nC_1x + {}^nC_2x^2 - .....$ ..... (2)

Multiply (1) and (2) and equality the coefficient of x^r on both sides

$\Rightarrow S_r$ = coefficient of x^r in $(1 - x^3)^n$

$$S_r = \begin{cases} 0 & \text{if } r \text{ is not a multiple of 3} \\ (-1)^{\frac{r}{3}} \cdot {}^nC_{\frac{r}{3}} & \text{if } r \text{ is a multiple of 3} \end{cases}$$

40. A, B

Sol. $|z_3| = \dfrac{|z_2||z_1 - z_4|}{|\bar{z}_1 z_4 - 1|} = \dfrac{|z_1 - z_4|}{|\bar{z}_1 z_4 - 1|} \le 1$; $|z_1 - z_4| \le |\bar{z}_1 z_4 - 1|$

$\Rightarrow |z_1 - z_4|^2 < |\bar{z}_1 z_4 - 1|^2 \Rightarrow |z_1|^2 + |z_4|^2 - |z_1|^2|z_4|^2 - 1 \le 0$

$\left(|z_1|^2 - 1\right)\left(|z_4|^2 - 1\right) \ge 0 \;\therefore\; |z_4| \ge 1$

41. A, C, D

Sol. Consider $\vec{a} = 6\hat{i} + 3\hat{j} + 2\hat{k}$; $\vec{b} = \left(\dfrac{s-a}{6}\right)\hat{i} + \left(\dfrac{s-b}{3}\right)\hat{j} + \left(\dfrac{s-c}{2}\right)\hat{k}$

$\Rightarrow \left(\vec{a} \cdot \vec{b}\right)^2 \le |\vec{a}|^2|\vec{b}|^2 \Rightarrow \dfrac{(s-a)^2}{36} + \dfrac{(s-b)^2}{9} + \dfrac{(s-c)^2}{4} \ge \dfrac{s^2}{49}$

$\Rightarrow \dfrac{s-a}{36} = \dfrac{s-b}{9} = \dfrac{s-c}{4} = \dfrac{s}{49}$; a = 13, b = 40, c = 45

42. A, D

Sol. Put x = x – 10 in f(10 + x) = f(10 – x) $\Rightarrow$ f(x) = f(20 – x)

From 2nd equation f(20 + x) = –f(20 – x)

$\Rightarrow$ f(x) = –f(20 + x) $\Rightarrow$ f(x + 20) = –f(40 + x)

$\Rightarrow$ –f(x) = –f(40 + x) $\Rightarrow$ f(x) = f(x + 40)

43. A, C, D

Sol. $\because BB^T = I, B^{-1} = B \Rightarrow B^T = B^{-1} = B$ and $B^2 = I, A^2 = I$

$X = \text{adj}(AB^T)^n = \text{adj}(AB)^n = \text{adj}(A^nB^n)$ [$\because$ A and B commute]

$= \text{adj } B^n \cdot \text{adj } A^n$

(A) If n is even $\Rightarrow A^n = B^n = I \Rightarrow X = I$

(B) If n is odd $\Rightarrow A^n = A, B^n = B$

$\Rightarrow X = \text{adj } B \cdot \text{adj } A = \text{adj}(AB) = (AB)^{-1}(|AB|) = B^{-1} A^{-1} = BA$ ($\because$ |AB| = 1 > 0)

44. B, C, D

Sol. Clearly x and y must be positive for S_1 and S_2

So, equation of $S_1 = x^2 + y^2 = 1$ and equation of S_2 : xy = 1

45. B

46. A

Sol. **(for Q. 45.-46):**

Let $A(x_1, y_1)$ and $B(x_2, y_2)$ where $F(1, 0)$, then $x_1 = \frac{y_1^2}{4}$ and $x_2 = \frac{y_1^2}{4}$

$\Rightarrow x_1x_2 + y_1y_2 = -4 \Rightarrow \frac{1}{16}(y_1y_2)^2 + y_1y_2 = -4 \Rightarrow \frac{(y_1y_2+8)^2}{16} = 0 \Rightarrow y_1y_2 = -8$

Area of ΔOFA = Area of ΔOFB = $\left(\frac{1}{2}|OF||y_1|\right)\left(\frac{1}{2}|OF||y_2|\right) = \frac{1}{4}|OF|^2 \cdot |y_2y_2| = 2$

$x_1x_2 = 4$

$FA = 1 + x_1$; $FB = 1 + x_2$

$FA + FB \geq 1 + x_1 + 1 + x_2 = 2 + x_1 + x_2$

$\frac{x_1 + x_2}{2} \geq \sqrt{x_1x_2}$ by AM – GM inequality

So, $FA + FB \geq 6$

Also, $FA \cdot FB = (1 + x_1)(1 + x_2) = 1 + x_1 + x_2 + (x_1x_2) \geq 9$

47. B

48. A

Sol. **(for Q. 47.-48):**

Equation of tangent of parabola at $(t^2, 2t)$ is $ty = x + t^2$ ..... (1)

Equation of normal of ellipe $4x^2 + 5y^2 - 20 = 0$ at $\left(\sqrt{5}\cos\phi, 2\sin\phi\right)$ is

$2y\cos\phi = \sqrt{5}x\sin\phi - \sin\phi\cos\phi$ ..... (2)

From equation (1) and (2), we get $\frac{t}{2\cos\phi} = \frac{1}{\sqrt{5}\sin\phi} = \frac{t^2}{-\sin\phi\cos\phi}$

$\Rightarrow \cos\phi = 0$ or $-\frac{1}{\sqrt{5}} \Rightarrow \phi = 0$ or $\frac{3\pi}{2}$, $\pi - \cos^{-1}\left(\frac{1}{\sqrt{5}}\right)$ or $\pi + \cos^{-1}\left(\frac{1}{\sqrt{5}}\right)$

For $\phi = 0$ or $\frac{3\pi}{2} \Rightarrow t = 0$ so not acceptable $t = \frac{1}{\sqrt{5}}, -\frac{1}{\sqrt{5}}$ as $t = \frac{2\cos\phi}{\sqrt{5}\sin\phi}$

Section – B

49. 2025

Sol. By inclusion – exclusion principle, the required number of paths is

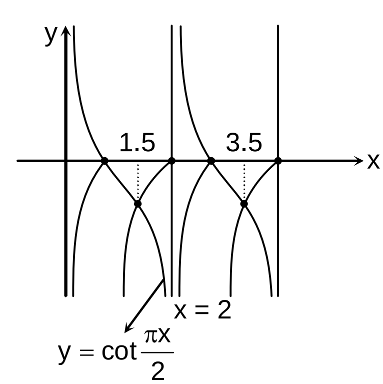
$= {}^{15}C_6 - {}^{6}C_3 \cdot {}^{9}C_3 - {}^{10}C_4 \cdot {}^{5}C_2 + {}^{6}C_3 \cdot {}^{4}C_1 \cdot {}^{5}C_2 = 2025$

50. 2525

Sol. Sum of solution = 1.5 + 3.5 + 7.5 + ... 99.5

= 2525

$y = \cot\frac{\pi x}{2}$ and $y = \log_2\{x\}$



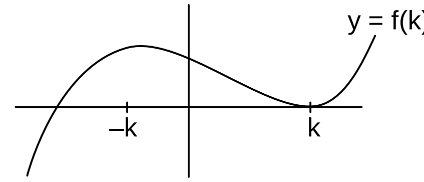
51. 108

Sol. $f'(x) = 3(x^2 - k^2) = 0$

For |x| = 0 have two distinct real roots

$f(k) = 0 \Rightarrow k = 2$

Required area = $\int_{-4}^{2} \left(x^3 - 12x + 16\right) dx = 108$ sq. unit



Section – C

52. 0.50

53. 4.00

Sol. **(for Q. 52.-53):**

$a = \frac{1}{2}$ and k = 4

$$\int_1^{\infty}\left(\frac{ax}{1+x^2} - \frac{1}{2x}\right)dx = \frac{a}{2}\log\left(1+x^2\right) - \frac{1}{2}(\log x)\Big|_1^{\infty} = \frac{1}{2}\log\frac{\left(1+x^2\right)^a}{x}\Bigg|_1^{\infty}$$

$$\Rightarrow \lim_{x\to\infty}\frac{1}{2}\log\frac{\left(1+x^2\right)^a}{x} - \frac{a\log 2}{2} \quad \therefore\ a = \frac{1}{2}$$

$$\int_1^{\infty}\left(\frac{ax}{1+x^2} - \frac{1}{2x}\right)dx = -\frac{\log 2}{4}$$

54. 4.00

55. 20.00

Sol. **(for Q. 54.-55):**

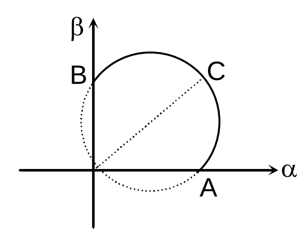
Let $\sqrt{y} = \alpha$, $\sqrt{x-y} = \beta$ $(\alpha, \beta \ge 0)$

Then, $x = y + \sqrt{x-y} = \alpha^2 + \beta^2$

Then equation is equivalent to $\alpha^2 + \beta^2 - 4\alpha = 2\beta$

$\Rightarrow (\alpha - 2)^2 + (\beta - 1)^2 = 5$ $(\alpha, \beta \ge 0)$

As seen in figure the trace of point (α, β) is the part of the circle with centre (2, 1) and radius $\sqrt{5}$



That is are ACB, then $\sqrt{\alpha^2 + \beta^2} \in \{0\} \cup \left[2, 2\sqrt{5}\right]$

$\Rightarrow x = \alpha^2 + \beta^2 \in \{0\} \cup [4, 20]$

56. 2.00

57. 11.00

Sol. **(for Q. 56.-57):**

Obviously f(0).f(1) > 0

$\Rightarrow 0 < c(a - b + c) \le \frac{a^2}{16} \Rightarrow c(a - b + c) \ge 1$ as a, b and c are positive integers equality holds when c = 1 and a = b

When c = 1, a = b, $b^2 - 4ac > 0$ implies $a^2 > 4a$, a > 4

$\Rightarrow$ So least integral value of a = 5, b = 5 and c = 1